Preface

The AI Explosion

- Scaling laws predict better models from faster hardware and bigger data, but the real surprise was the explosion of applications.
- Lowered entry barriers now allow developers to build applications without writing code, transforming AI into a universal tool.
- Adoption builds on decades-old techniques like language modeling (1950s) and retrieval systems, yet scale creates new challenges.

Core Engineering Principles

- Best practices from traditional ML—systematic experimentation, rigorous evaluation, and optimization—remain essential for foundation models.
- Familiarity can mislead engineers into thinking nothing is new, but increased model scale introduces unique opportunities and hurdles.

Core Purpose & Framework

- Provides a framework for adapting foundation models (LLMs and LMMs) to specific applications.
- Answers critical questions like "Should I build this?" and "How do I evaluate my application?".
- Covers best practices for prompt engineering, RAG strategies, agents, finetuning, and data validation.

Authoritative Fundamentals

- Draws on a decade of experience with language models and ML systems, illustrated by case studies.
- Focuses on three enduring fundamentals that outlast rapidly changing tools and APIs.
- Includes ample references and content extensively reviewed by experts from diverse

What This Book Is Not

- **Not a tutorial**: Offers a framework for selecting tools rather than teaching specific tool usage or pseudocode implementation.
- **Not an ML theory book**: Focuses on practical application building instead of explaining neural networks or training from scratch.
- **No prior ML required**: Readers can succeed without background, though understanding probabilistic concepts aids effectiveness.

Approach to Problem Solving

- **Start simple**: Analyzes fundamental challenges by progressing from simplest solutions to complex ones as needed.
- **Consult experts**: Relies on a network of researchers and engineers to identify important problems and solutions.
- **Apply Lindy's Law**: Infers future technology longevity based on current age, assuming established tech will persist longer.

Temporary Concepts Included

- **Immediate utility**: Includes concepts useful for specific application developers even if they are temporary.
- **Illustrative value**: Features concepts that demonstrate interesting problem-solving approaches despite potential obsolescence.

Target Audience Overview

This technical book targets AI engineers, ML engineers, data scientists, and technical product managers leveraging foundation models. It covers core concepts like supervision, gradient descent, backpropagation, loss functions, and hyperparameter tuning. Readers should be familiar with neural architectures (feedforward, recurrent, transformer) and metrics (accuracy, F1, cosine similarity).

Practical Application Scenarios

The book supports building or optimizing AI applications from scratch to production-ready stages. It addresses challenges like hallucinations, security, latency, and costs while offering targeted solutions. Readers can streamline team development processes to make them more systematic, faster, and reliable.

Strategic Business Impact

Navigating the Book Structure

- The book follows the typical AI application development process, from decision-making to end-to-end deployment.
- Modules cover skills for AI engineers, capabilities/limitations of AI, and technical deep dives.
- Readers can skip sections they are familiar with; early readers may find some technical details too dense.

Core Development Chapters Overview

- **Chapters 3 & 4**: Evaluate reliability and build systematic pipelines using different evaluation methods.
- **Chapter 5**: Master prompt engineering, best practices, and defenses against malicious prompt attacks.
- **Chapter 6**: Optimize responses using context construction patterns like RAG and agentic workflows.

Advanced Optimization & Deployment

- **Chapter 7**: Adapt models via finetuning and model merging while managing memory-intensive requirements.
- **Chapter 8**: Acquire, annotate, and process high-quality data essential for effective model training.
- **Chapter 9**: Optimize inference costs and speed at both the model and service levels; design user feedback systems.

Typographical Styles

- Italics indicate new terms, URLs, emails, filenames, and extensions.
- Constant width denotes program listings and code elements like variables.
- Bold constant width shows commands to be typed literally by users.
- Italicized constant width marks placeholders for user-supplied values.

Formatting Hierarchy

- **Bold**: Used for emphasis or critical actions requiring immediate attention.
- *Italic*: Highlights new terminology, web addresses, and file identifiers.
- `Monospace`: Displays code blocks and specific programmatic references.
- **`Monospace Bold`**: Designates commands that must be entered exactly as shown.

Markdown Elements

- **TIP**: Signifies a tip or suggestion for readers.
- **NOTE**: Indicates a general note or observation.
- **WARNING**: Alerts users to potential risks or cautions.
- These elements guide reader attention and provide context.

Code Example Resources

- Supplemental code, exercises, and papers are hosted on GitHub.
- The repository covers deep AI engineering topics beyond the book's scope.
- It includes behind-the-scenes writing process information and statistics.
- Technical support for code examples is available via email.

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- The web page lists updates and additional information regarding the book's content.

Key Contributors & Technical Reviewers

- Luke Metz and Han-chung Lee provided early technical reviews and prevented wrong paths.
- Vittorio Cretella and Andrei Lopatenko offered deep expertise from Fortune 500 AI leadership.
- Eugene Yan, Shawn Wang, and Sanyam Bhutani gave critical technical and emotional support.

Editorial Team & Career Lessons

- Development editors Melissa Potter, Corbin Collins, and Jill Leonard shaped the book's structure.
- Production editor Elizabeth Kelly and copyeditor Liz Wheeler ensured quality production standards.
- The author learned that tools become outdated quickly and that better language models are essential.